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Effect of different storage conditions on β-carotene retention in two varieties of golden sweet potato (*Ipomoea batatas*)

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Abstract- The present study evaluates the effect of diverse storage conditions-sunlight, shade, and normal ambient temperature-on the retention of β-carotene in two varieties of Golden Sweet Potato (*Ipomoea batatas*), namely CIP-440107 and ST-14. Freshly harvested tubers were stored for 15, 30, and 45 days, and β-carotene concentrations were quantified to determine varietal responses to environmental stress. Results revealed pronounced varietal differences in carotenoid stability. CIP-440107 exhibited substantial β-carotene degradation under sunlight, with consistent losses of approximately 56% across all storage durations, indicating high sensitivity to photo-oxidative and thermal stress. Shade emerged as the most effective storage condition for this variety, limiting β-carotene loss to only 6-8%, while ambient storage caused moderate declines (13-15%). In contrast, ST-14 showed exceptional tolerance to sunlight, maintaining nearly stable β-carotene levels throughout the storage period, but demonstrated higher susceptibility to degradation under normal indoor conditions (17-19% loss). These contrasting patterns demonstrate the importance of varietal physiological characteristics in determining carotenoid stability. The findings emphasize the need for variety-specific post-harvest storage recommendations to ensure nutritional preservation, optimize β-carotene retention, and enhance the practical utility of biofortified Golden sweet potato in nutrition-sensitive agricultural systems.

Keywords: β-carotene retention, Golden sweet potato, storage conditions, carotenoid stability, varietal response, photo-oxidation

INTRODUCTION

Golden Sweet potato (*Ipomoea batatas* L.) is globally recognized as a nutritionally dense, climate-resilient tuber crop valued for its high carbohydrate content, dietary fiber, minerals, and provitamin A carotenoids.[1] Among its pigments, β-carotene is the most significant because it serves as a precursor of vitamin A and contributes to the orange coloration of the flesh. Orange-fleshed sweet potato (OFSP) varieties have consequently received attention in biofortification programs aimed at combating vitamin A deficiency in low-income regions.[2] The rise of consumer interest in functional foods has further amplified the importance of OFSP as a biofortified crop with superior nutritional and health-promoting qualities.[3] However, despite their nutritional value, Golden sweet potatoes are highly perishable and undergo rapid physiological deterioration if not stored under appropriate environmental conditions.

Storage environment significantly affects the biochemical quality, physical attributes, textural stability, and post-harvest longevity of Golden sweet potato tubers. Conditions such as sunlight exposure, ambient temperature, shade, humidity, and aeration regulate moisture loss,

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respiration rate, sprouting, and the degradation of carotenoids.[4] Prolonged exposure to direct sunlight accelerates transpiration, increases respiration, and causes dehydration and hardening of tubers, which ultimately leads to shrinkage, weight loss, and compromised consumer acceptability. In contrast, shade storage minimizes thermal stress and reduces moisture loss, allowing tubers to maintain relatively high quality for longer durations.[5] Normal room-temperature storage, while commonly practiced, may cause sprouting and moderate physiological deterioration depending on humidity and ventilation. Understanding these storage-induced changes is essential for optimizing the shelf life and nutritional integrity of Golden sweet potato varieties cultivated in local farming systems.

The Golden Sweet Potato, increasingly cultivated in parts of India, demonstrates considerable promise due to its vibrant coloration and nutritional attributes. In this study, two varieties of Golden Sweet Potato were evaluated: (1) CIP – 440127, which possesses a white outer covering with orange flesh and a relatively low β-carotene concentration, and (2) ST-14, which has a reddish outer covering, orange flesh, and significantly higher β-carotene levels.

Colour changes during storage are also critical indicators of carotenoid stability. β-carotene is the primary determinant of the orange hue in Golden sweet potato flesh; thus, carotenoid degradation often manifests as fading or dulling of color.[6] Direct sunlight enhances surface heating and accelerates the rate of moisture evaporation, leading to excessive shrinkage and hardening of tubers. This phenomenon can be especially pronounced after 45 days of sunlight exposure, as reported in many studies documenting extreme textural degradation under harsh storage conditions.[7] In contrast, shade storage reduces radiant heat and slows moisture loss, allowing tubers to retain smoother texture, color vibrancy, and structural integrity.

MATERIAL & METHOD

Two varieties of Golden Sweet Potato (*Ipomoea batatas* L.) were selected for the study, CIP – 440127 variety and ST-14 variety. Freshly harvested disease-free tubers of uniform size (100-150 g), free from cuts, bruises, or mechanical injury, were procured directly from farmers in the study region. All samples were washed thoroughly to remove soil particles and air-dried before storage. The storage conditions for the experiment were categorized into three distinct environments. In the sunlight condition (S_1), tubers were placed in an open area where they received direct sunlight for 6–7 hours per day, with mean daytime temperatures ranging between 30–36°C. The shade condition (S_2) involved storing the tubers under a well-ventilated shaded structure that prevented direct exposure to sunlight while allowing free airflow, maintaining temperatures between 24–30°C. In contrast, the normal room storage condition (S_3) consisted of keeping the tubers indoors at ambient temperatures of 25–32°C, with moderate natural light and no forced ventilation. These three environments provided distinct thermal and light conditions, enabling a comparative assessment of their effects on β-carotene retention in Golden sweet potato tubers.

Standard preparation

The standard stock solution at 1 mg/ml concentration was prepared by dissolving standard in acetone. The working standard solutions of 32, 16, 8, 4, 2, 1, 0.50, 0.25, 0.125, 0.062, and 0.03 mg/ml were prepared. A working standard containing 32 mg/ ml was prepared from the 1 mg/ml stock solutions kept at 4°C. From this working standard different dilutions were made to spike the samples. Blank samples of 1.0 g were spiked with working standards to obtain final concentrations 16.0, 8.0, 4.0, 2.0, 1.0, 0.5, 0.25, 0.125, 0.062, 0.031 and 0.015 mg/g of b- carotene and extracted as described previously. Calibration curves were plotted by taking Optical Density (O. D.) value to the respective concentrations by back extrapolation methods. These curves were used to quantify the b-carotene content in the samples analyzed.

Estimation of β-carotenoid content

The estimation of β-carotenoid content done by the spectrophotometric method of Schierle, Schellenberger, Fizet, & Betz, 2002 and the acetone was used as extractant. For extraction, a representative portion of this sample (1 g) was accurately weighed in a glass test tube. Then 5 ml of chilled acetone was added to it, and the tube was held for 15 min with occasional shaking at 4-1°C, vortexed at high speed for 10 min, and finally centrifuged at 1370 g for 10 min. Supernatant was collected into a separate test tube, and the compound was re-extracted with 5 ml of an acetone followed by centrifugation once again as above. Both of the supernatants were pooled together and then passed through the Whatman filter paper No. 42. The absorbance of the extract was determined at 449 nm wavelength in a UV Vis spectrophotometer.

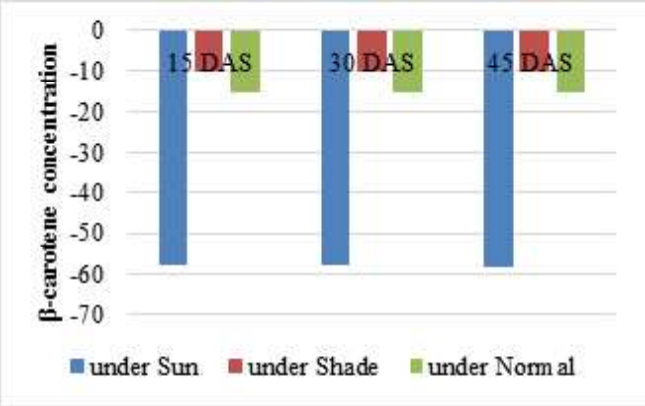
RESULT

Effect of Storage Environment on β-Carotene Retention (%) in Golden Sweet Potato Variety CIP – 440127-441107

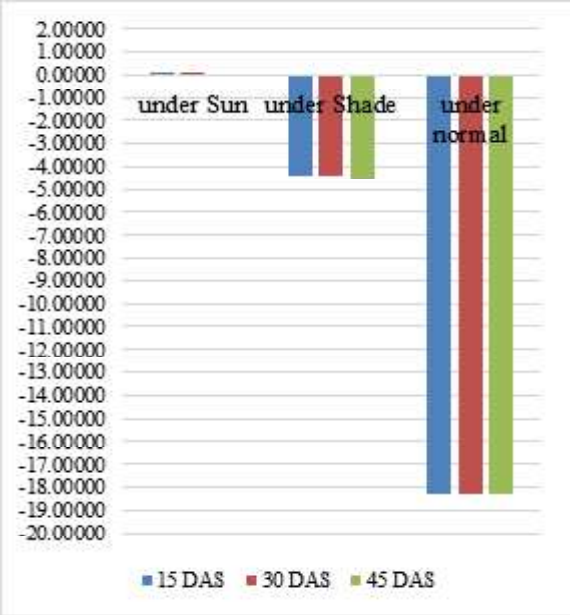
Under sunlight storage conditions, the β-carotene concentration exhibited the highest rate of degradation, with a consistent loss of approximately 56% recorded at 15, 30, and 45 days after storage (DAS). This sharp decline within the first 15 days indicates that more than half of the total β-carotene content was rapidly destroyed, primarily due to intense photo-oxidation and elevated temperatures. The lack of further significant deterioration after 15 days suggests that most of the susceptible carotenoid fraction had already degraded early during exposure to direct sunlight. In contrast, storage under shade resulted in minimal loss of β-carotene, with reductions of about 10% at 15 DAS, 9% at 30 DAS, and 11% at 45 DAS. These values indicate that approximately 89–91% of the β-carotene content was retained throughout the storage period, demonstrating that shade conditions provide effective protection by limiting light exposure and maintaining relatively lower temperatures, thus reducing oxidative damage. Meanwhile, normal room storage conditions showed moderate β-carotene degradation, with losses ranging between 15% and 16% across all time points. This indicates that 84–85% of the carotenoid content remained preserved, representing a level of protection superior to sunlight but inferior to shaded conditions. The moderate and steady decline under ambient conditions is likely due to intermittent exposure to light, temperature fluctuations, and gradual moisture loss, all of which contribute to slow but continuous oxidative breakdown of β-carotene.

Effect of Storage Environment on β-Carotene Retention (%) in Golden Sweet Potato Variety ST-14

The β-carotene concentration of variety ST-14 shows clear variation across storage conditions and days after storage (DAS), when expressed in percentage loss. Under sunlight, the values remain around -3 to -4% at 15, 30, and 45 DAS, indicating a small but consistent decline. Although the reduction appears minor, the uniformity across days suggests that most β-carotene degradation happens early, with sunlight continuously promoting low-grade pigment oxidation. Under shade, the losses remain within -5 to -6%, showing a slightly higher reduction compared to sun but still relatively minimal. This indicates that shade storage slows photo-oxidation effectively, allowing the tubers to retain approximately 94–95% of their β-carotene even after 45 days. In contrast, normal storage conditions show the most substantial decline, with β-carotene losses reaching approximately -18 to -19% across 15, 30, and 45 DAS. This means only about 81–82% of the pigment remains under ambient conditions. The relatively large and stable reduction indicates that fluctuating room temperatures, mild light exposure, and moisture loss promote steady oxidative degradation. Overall, shade storage provides the best preservation of β-carotene in variety ST-14, while normal room storage results in the highest loss.



Graph 1 – Showing β-carotene concentration under different storage condition of variety CIP – 440127-440107 after 15, 30 and 45 DAS (Days After Storage)



Graph 2 – Showing β-carotene concentration under different storage condition of variety ST-14 after 15, 30 and 45 DAS (Days After Storage)

Table 1- Showing Mean± St. deviation values (OD at 449) of chips packet samples after 15, 30 and 45 days of Storage kept under Sun and Shadow.

Type of packet material	Days after packaging	Mean± St. deviation		
CIP – 440127 - 440107		Under sun	Under shadow	Normal
	15 DAP	0.321±0.016	0.376 ± 0.012	0.132 ± 0.001
	30 DAP	0.265±0.002	0.314 ± 0.006	0.126 ± 0.003
	45 DAP	0.085± 0.008	0.299 ± 0.002	0.114 ± 0.002
ST-14	15 DAP	0.526±0.006	0.3 ±0.012	0.131 ± 0.007
	30 DAP	0.476±0.006	0.246 ± 0.005	0.123 ± 0.008
	45 DAP	0.325±0.003	0.131 ± 0.004	0.116 ± 0.005



Fig. 1 Field Visit by Scientists to the Golden Sweet Potato Cultivation Area

DISCUSSION

Bechoff *et al.* (2010)[8] explained that β-carotene breakdown in Golden sweet potato chips follows temperature-dependent degradation kinetics, reinforcing the conclusion that heat-intensive environments, such as open sun storage, promote pigment loss. Light-induced deterioration has also been reported; Kato *et al.* (2024)[9] found that prolonged exposure to sunlight triggers oxidative stress in stored Golden sweet potato roots, leading to altered pigment composition, consistent with the patterns of β-carotene depletion observed in sensitive varieties in this study. In addition, a more recent study by Ramirez-Hernandez *et al.* (2025)[10] highlighted that combined heat and photo-oxidative stress can significantly accelerate carotenoid isomerization and breakdown in orange-fleshed root crops, providing further evidence that sun-exposed storage environments present a high-risk condition for β-carotene destabilization

CONCLUSION

Golden sweet potato is highly valued for its rich β-carotene content, a powerful precursor of vitamin A essential for human health. β-carotene plays a critical role in maintaining good vision, especially night vision, by supporting the formation of visual pigments in the retina. Storage conditions play a decisive role in determining the quality, nutritional value, and β-carotene stability of Golden Sweet Potato. Across all parameters measured, sunlight exposure proved to be the most detrimental storage condition, resulting in the highest weight loss, extreme shrinkage, excessive hardening, and rapid fading of flesh colour. These effects were especially severe after 45 days, reflecting accelerated moisture loss and oxidative degradation of pigments, including β-carotene. Both varieties showed significant deterioration under sunlight, highlighting that direct solar radiation is unsuitable for extending Golden sweet potato shelf life. Varietal differences were also evident. The ST-14 variety, with its higher β-carotene concentration, showed greater sensitivity to sunlight-induced pigment degradation but retained superior color when stored under shade. Meanwhile, the CIP - 440127 variety, with lower β-carotene levels, demonstrated slightly better tolerance to sunlight exposure but did not match the color intensity or nutritional potential of ST-14 under optimal conditions. Overall, the study concludes that shade storage is the most appropriate and scientifically supported method for maintaining the post-harvest quality of Golden Sweet Potato, especially for β-carotene-rich varieties such as ST-14. These findings have practical significance for farmers, traders, and storage managers seeking to reduce post-harvest losses and preserve the nutritional benefits of Golden sweet potato.

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